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Vellore Institute of Technology

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SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS

(SCORE)

M.TECH [S.E]

WINTER SEMESTER 2025-2026

SWE2009- DATA MINING TECHNIQUES

J COMPONENT

REVIEW 3

TITLE: CROP DISEASE PREDICTION

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1. ABSTRACT

Crop diseases pose a serious threat to agricultural productivity, economic stability, and global food security. Early and accurate identification of crop diseases is essential to minimize yield loss, reduce excessive pesticide usage, and support sustainable farming practices. This project aims to develop an efficient and automated crop disease prediction system using machine learning techniques. A labeled dataset consisting of images of various crop diseases was collected and preprocessed through resizing, normalization, and data augmentation to enhance model performance. Multiple classification algorithms, including Random Forest, Support Vector Machine (SVM), Decision Tree, Naive Bayes, and XG Boost, were implemented and evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score.

Comparative analysis revealed that the XG Boost model outperformed other classifiers due to its ability to handle complex feature interactions and reduce overfitting through gradient boosting. Consequently, XG Boost was selected for final testing and validation. The testing phase involved predicting disease classes from randomly selected images across all categories, where the model achieved an accuracy of 86.4%. The results highlight the robustness and reliability of XG Boost for image-based crop disease classification. This system demonstrates strong potential for real-time deployment in smart agriculture applications, enabling early disease detection, informed decision-making, and improved crop management. Future enhancements may include integration with IoT devices, mobile-based farmer assistance systems, and deep learning models for further accuracy improvements.

2. INTRODUCTION

Agriculture plays a vital role in ensuring food security and sustaining the livelihoods of millions of people worldwide. However, crop diseases continue to be a major challenge in agricultural production, often leading to significant yield loss and economic damage when not detected at an early stage. Timely identification of plant diseases is essential to prevent their spread and to reduce excessive use of pesticides, which can harm both the environment and human health. Traditional methods of crop disease detection rely heavily on manual inspection by agricultural experts, which is time-consuming, labor-intensive, and often impractical for large-scale farming and continuous monitoring.

Recent advancements in machine learning and computer vision have introduced efficient and automated solutions for crop disease detection. Image-based disease identification systems enable accurate analysis of plant health by learning patterns from visual symptoms present on leaves. These technologies provide a scalable and cost-effective alternative to conventional methods and support farmers in making informed decisions at the right time.

In this project, a crop disease prediction system is developed using the Paddy Disease Dataset, which consists of 1,850 labeled images of paddy leaves classified into six categories: Bacterial Leaf Blight, Brown Spot, Healthy Leaf, Leaf Blast, Others, and Tungro. The primary objective of the study is to accurately identify and classify paddy leaf diseases using machine learning techniques, thereby improving early disease detection and crop management.

The proposed methodology involves image preprocessing steps such as resizing and normalization, followed by training multiple machine learning classifiers including Random Forest, Support Vector Machine (SVM), Decision Tree, Naive Bayes, and XGBoost. These models are evaluated using standard performance metrics, and XGBoost is identified as the best-performing algorithm due to its robustness and superior predictive capability. The final model achieves a testing accuracy of 86.4% on unseen data. The results of this study demonstrate the effectiveness of machine learning-based crop disease prediction systems and highlight their potential for real-world deployment in smart agriculture applications, contributing to timely disease management and improved crop productivity.

3. LITERATURE REVIEW

1. Title – A Comprehensive Review of Machine Learning-Based Approaches to Detect Crop Diseases.

Authors: Rajesh Kumar & Vikram Singh

Publisher/Journal: ResearchGate

Year: 2023

Summary

This paper presents a comprehensive review of machine learning–based techniques used for crop disease detection through image analysis. It discusses how agricultural images are collected and preprocessed before applying classification algorithms to identify diseases in crops such as rice, wheat, tomato, and cotton. The study mainly focuses on traditional machine learning approaches including feature extraction, pattern recognition, and supervised classification methods. Various stages of the disease detection pipeline, such as image segmentation and feature selection, are explained in detail. Although the paper provides a broad overview of existing ML techniques and their effectiveness, it offers limited discussion on advanced deep learning models. Overall, the review highlights the importance of machine learning in automated crop disease detection while emphasizing the need for more scalable, crop-independent, and real-world applicable solutions.

Data Mining Techniques and Algorithms Used

- Feature extraction using color, texture, and shape features
- Use of classifiers like SVM, Random Forest, and Decision Tree
- Image preprocessing techniques like segmentation and filtering
- Pattern recognition for identifying disease symptoms

Research Gap Identified

- Limited use of deep learning models
- Lack of real-world dataset validation
- Models are not scalable for multiple crops
- Less focus on early disease detection

Future Enhancements / Scope for Improvement

- Use deep learning models like CNN
- Include real-time farm and sensor data
- Develop models for multiple crop types
- Improve early-stage disease detection

2. Title – A Systematic Analysis of Machine Learning and Deep Learning Based Approaches for Identifying Plant Diseases

Authors: Imtiaz Ahmed & Pramod K. Yadav

Publisher/Journal: ScienceDirect

Year : 2023

Summary

This paper provides a systematic analysis of machine learning (ML) and deep learning (DL) approaches used for plant disease identification. It compares traditional ML techniques that rely on manual feature extraction with deep learning models that automatically learn features from images. The study highlights popular datasets such as PlantVillage that are widely used for training and evaluating disease detection models. Various algorithms are examined in terms of accuracy, feature dependency, and computational complexity. The paper demonstrates that deep learning models, particularly CNNs, generally outperform classical ML approaches.
However, it also notes challenges related to dataset diversity and real-world applicability. Overall, the paper offers valuable insights into the strengths and limitations of ML and DL methods and emphasizes the need for more generalized and scalable disease detection solutions.

Data Mining Techniques and Algorithms Used

- Support Vector Machine (SVM) and Random Forest for classification
- Convolutional Neural Networks (CNN) for automatic feature learning
- Transfer Learning models (VGG, ResNet)
- Image segmentation and preprocessing techniques

Research Gap Identified

- Over-reliance on PlantVillage benchmark dataset
- High computational cost of deep learning models
- Lack of evaluation under real farm conditions

Future Enhancements / Scope for Improvement

- Development of lightweight DL models for mobile applications
- Combining image data with environmental and soil parameters
- Adoption of real-time disease monitoring systems

3. Title – Construction of deep learning-based disease detection model in plants

Authors: Minah Jung, Jong Seob Song, Ah Young Shin, Beomjo Choi, Sangjin Go, Suk Yoon Kwon, Juhan Park,Sung Goo Park & Yong Min Kim.

Publisher/Journal: Scientific reports (Nature)

Year: 2023

Summary

This paper proposes a deep learning-based framework for automatic plant disease detection using leaf images. The model employs multiple pre-trained convolutional neural networks (CNNs) in a stepwise classification process to first identify crop type and then detect and classify plant diseases. Image preprocessing and data augmentation techniques are used to improve model accuracy and robustness. Experimental results show high classification accuracy, demonstrating the effectiveness of deep learning for plant disease detection. However, the evaluation is mainly limited to selected crop categories, which restricts the model's general applicability. The study highlights the potential of CNN-based architectures while emphasizing the need for broader datasets and generalized frameworks for real-world agricultural deployment.

Data Mining Techniques and Algorithms Used

- Utilized CNN-based deep learning models with pre-trained architectures such as VGG19, ResNet50, EfficientNet, and AlexNet for plant disease detection.

- Adopted a multi-stage classification framework: crop identification → disease detection → disease classification.
- Employed image preprocessing techniques including resizing, normalization, and augmentation to improve model robustness.

Research Gap Identified

- The study focuses on specific crops (mainly Solanaceae family), limiting generalization across diverse crop varieties.
- The system does not provide a universal crop-independent disease detection model.
- Disease detection is mainly performed at visible symptom stages, with limited emphasis on early-stage disease identification.

Future Enhancements / Scope for Improvement

- Development of a crop-independent deep learning framework applicable to diverse plant species.
- Incorporation of ensemble learning methods combining multiple CNN models to increase accuracy and robustness.
- Extension of the model to support early-stage disease detection and multi-severity disease classification under real-field conditions.

4. Title – Plant Disease Detection Using 2D CNN Based Deep Learning Architecture

Authors: P. Dharmendra Kumar, A. Suhasini, D.Anand

Publisher/Journal: IJISAE

Year: 2023

Summary

This paper proposes a 2D Convolutional Neural Network (CNN) architecture for automatic detection of plant diseases using leaf images. The approach applies image preprocessing techniques such as grayscale conversion, resizing, and noise reduction before classification. Supervised learning is used to train the model on labeled datasets, enabling effective identification of different disease categories. The CNN automatically extracts and selects relevant features, improving classification efficiency and reducing manual feature engineering. Experimental results show reliable performance; however, the evaluation is restricted to limited

datasets and specific crops. The model mainly focuses on clearly visible disease symptoms, making it less effective for early disease detection. Overall, the study demonstrates the feasibility of CNN-based approaches while highlighting the need for scalable and generalized disease detection systems.

Data Mining Techniques and Algorithms Used

- Proposed a 2D Convolutional Neural Network (CNN) for automated plant leaf disease classification.
- Applied standard image preprocessing techniques such as grayscale conversion, resizing, and noise reduction.
- Used supervised learning with labeled datasets for training and validation.
- The CNN model implicitly performed feature selection and extraction, improving classification efficiency.

Research Gap Identified

- The model was tested on limited datasets and crop types, restricting scalability.
- Absence of a generalized disease detection architecture suitable for multiple crops.
- Limited analysis on early disease symptoms, focusing mostly on clearly infected leaf images.

Future Enhancements / Scope for Improvement

- Integration of ensemble CNN models to improve detection accuracy and reduce misclassification.
- Expansion towards a universal disease detection system supporting multiple crop varieties.
- Enhancement of the framework to detect early-stage and multiple disease stages, enabling timely intervention for farmers.

5. Title – Unsupervised Deep Learning Techniques for Automatic Detection of Plant Diseases

Authors: Alessandro Benfenati, Paola Causin, Roberto Oberti & Giovanni Stefanello

Publisher/Journal: Springer

Year: 2023

Summary

This paper introduces unsupervised deep learning techniques for automatic plant disease detection without relying on manually labeled datasets. The study primarily focuses on cucumber leaf disease detection and demonstrates how clustering-based learning methods can identify infected regions automatically. CNN-based feature extraction is combined with unsupervised clustering algorithms to differentiate healthy and diseased areas. Dimensionality reduction techniques are used to enhance computational efficiency and feature representation. The approach significantly reduces the dependency on labeled data, which is a major challenge in agricultural applications. However, the framework is validated on a single crop type and lacks extensive real-field evaluation. The study highlights the potential of unsupervised learning while emphasizing the need for broader validation and hybrid learning strategies.

Data Mining Techniques and Algorithms Used

- Unsupervised deep learning for disease detection
- CNN-based automatic feature extraction
- Clustering algorithms for disease region identification
- Dimensionality reduction techniques

Research Gap Identified

- Applied to a single crop and disease type
- Limited validation using real field datasets
- Lack of comparison with supervised learning approaches

Future Enhancements / Scope for Improvement

- Extension to multi-crop and multi-disease detection
- Combination of unsupervised and semi-supervised learning
- Use of multispectral and hyperspectral image data

6. Title: Exploration of Machine Learning Approaches for Automated Crop Disease Detection

Authors: panelAnnu Singla a 1, Ashima Nehra b 1, Kamaldeep Joshi a, Ajit

Kumar c, Narendra Tu teja d, Rajeev K. Varshney e, Sarvajeet Singh Gill b, Ritu Gill

Year: 2024

Publisher: Journal in Agricultural Informatics and Machine Learning (Peer-

Reviewed)

Summary

This paper provides a detailed analysis of machine learning and deep learning techniques for automated crop disease detection. Traditional disease identification methods primarily depend on manual inspection by experts, which is time-consuming, labor-intensive, and often leads to inaccurate diagnosis due to human error and environmental variations. To overcome these limitations, the authors explore advanced machine learning models such as Convolutional Neural Networks (CNN), Support Vector Machines (SVM), and Random Forest classifiers, along with deep learning architectures including ResNet and Inception. The study emphasizes the importance of image preprocessing, feature extraction, and robust evaluation metrics in improving classification accuracy. By automating disease detection, the proposed approaches support early diagnosis, reduce crop loss, and contribute to sustainable agricultural practices and food security.

Data Mining / Machine Learning Techniques & Algorithms Used

The study employs a combination of supervised machine learning and deep learning algorithms, including CNN, SVM, Random Forest, and Decision Trees. Advanced deep neural networks such as ResNet and DenseNet are used for learning complex image features. Feature extraction and dimensionality reduction techniques like Principal Component Analysis (PCA) are applied to reduce computational complexity. Image preprocessing and data augmentation techniques further enhance model performance and generalization.

Research Gap Identified

Despite achieving high prediction accuracy, the models require large volumes of high-quality labeled datasets, which are difficult to obtain in real agricultural environments. The computational cost associated with deep learning models limits their deployment on lowresource devices. Moreover, generalization across different crops, geographical regions,

and environmental conditions remains a major challenge. The lack of explainability in deep learning models also reduces farmer trust and practical adoption.

Future Enhancements / Scope for Improvement

Future research can focus on developing lightweight, energy-efficient, and explainable models suitable for real-time deployment on mobile and edge devices. Integration with IoT sensors and edge computing platforms can improve scalability and real-world usability. Expanding datasets to include diverse crop varieties, climatic conditions, and disease stages will further enhance robustness and generalization.

7. Title: An Approach for Crop Prediction in Agriculture: Integrating Genetic Algorithms and Machine Learning

Authors: Tanjim Mahmud, Nippon Datta, Rishita Chakma, Utpol Kanti Das,

Mohammad Tarek Aziz, Musaddikul Islam, Abul Hasnat Muhammed Salimullah,

MohammadShahadatHossain,KarlAndersson

Year: 2024

Publisher: IEEE Xplore

Summary

This paper proposes an intelligent crop prediction framework that integrates Genetic Algorithms (GA) with machine learning techniques to support data-driven agricultural decision-making. The system uses soil characteristics and weather parameters to recommend suitable crops for cultivation. Genetic Algorithms are employed to optimize feature selection and model parameters, improving prediction accuracy and reducing redundancy in input data. A Random Forest classifier is used for final crop prediction, achieving high accuracy across multiple crop classes. The integration of Explainable AI techniques such as SHAP and LIME enhances transparency, allowing farmers and stakeholders to understand model decisions and optimize resource utilization for smart agriculture.

Data Mining / Machine Learning Techniques Used

The proposed approach combines Genetic Algorithms for feature and parameter optimization with a Random Forest classifier for crop prediction. The Random Forest model consists of 86 decision trees with a maximum depth of 59 and achieves a prediction accuracy of 99.3% across

22 crop classes. Explainable AI methods, including SHAP and LIME, are used to analyze feature importance and improve interpretability.

Research Gap Identified

Although the proposed model demonstrates excellent accuracy, it is computationally expensive due to the use of Genetic Algorithms and explainability techniques such as SHAP. The complex model structure increases the risk of overfitting, especially when trained on high-dimensional data. Additionally, real-time deployment may be challenging in regions with limited computational infrastructure.

Future Enhancements / Scope for Improvement

Future studies can focus on reducing computational overhead and improving efficiency for real-time applications. Hybrid optimization techniques and lightweight explainability methods can be explored to enhance scalability. Incorporating real-time IoT-based data and extending the framework to diverse climatic regions and crop varieties will further improve applicability**.**

8. Title: A Deep Learning-Based Crop Disease Diagnosis Method Using Multimodal Mixup Augmentation

Authors: Hyunseok Lee, Young-Sang Park, Songho Yang, Hoyul Lee, Tae-JinPark,DoyeobYeo

Year: 2024

Publisher: MDPI – Applied Sciences

Summary

This paper introduces a deep learning-based crop disease diagnosis framework that integrates image data with environmental time-series data to enhance prediction accuracy. The authors combine a CNN-based ResNet-50 model for image feature extraction with an LSTM network to analyze environmental parameters such as temperature and humidity. A novel multimodal mixup augmentation technique is proposed to improve data fusion and model generalization. Experimental results show that the proposed method significantly outperforms traditional image-only approaches, making it suitable for smart agriculture and real-world disease monitoring.

Data Mining / Machine Learning Techniques Used

The proposed approach employs advanced deep learning and data mining techniques to improve crop disease diagnosis by integrating multimodal data. A Convolutional Neural Network (CNN) based on the ResNet-50 architecture is used for automatic feature extraction from crop leaf images, capturing complex visual patterns related to disease symptoms. To analyze environmental time-series data such as temperature and humidity, a Long Short-Term Memory (LSTM) network is utilized due to its ability to model temporal dependencies. A novel multimodal mix up augmentation technique is applied to fuse image features and environmental data, enhancing model generalization and robustness. The model training process is optimized using the Adam optimizer with an adaptive learning rate. Performance evaluation is carried out using standard classification metrics including accuracy, precision, recall, and F1-score.

Research Gap Identified

The availability of large-scale multimodal datasets remains limited, restricting extensive validation. Environmental sensor data may also introduce noise, affecting prediction reliability under real-world conditions.

Future Enhancements / Scope for Improvement

Future work can focus on collecting diverse multimodal datasets, improving sensor reliability, and integrating IoT-based real-time monitoring systems for large-scale deployment.

9. Title: Enhancing Crop Yield Predictions with PEnsemble 4: IoT and MLDriven for Precision Agriculture

Authors: Kiattisak Sangpradit, Attaphongse Taparugssanagorn, Nisit Pukrongta

Year: 2024

Publisher: MDPI – Applied Sciences

Summary

This paper presents PEnsemble 4, an advanced machine learning framework designed to enhance crop yield prediction in precision agriculture. The model integrates IoT sensor data and UAV-based imagery to predict maize yield at an early growth stage (R2 stage). Environmental parameters such as temperature, humidity, and soil characteristics are combined with vegetation indices extracted from UAV images to enable early and proactive agricultural decision-making. The weighted ensemble approach achieves a prediction accuracy of 91%,

outperforming individual machine learning and deep learning models and supporting datadriven farming practices.

Research Gap Identified

The proposed approach depends heavily on high-quality IoT sensor data and UAV imagery, which may not be accessible to all farmers. The computational cost associated with processing large-scale sensor and image data limits deployment in resource-constrained environments.

Future Enhancements / Scope for Improvement

Future research can focus on optimizing computational efficiency, reducing infrastructure costs, and developing cloud-edge hybrid solutions. Extending the framework to multiple crops and geographic regions will further enhance generalizability and adoption.

10. Title: Deep Learning and Content-Based Filtering Techniques for Improving Plant Disease Identification and Treatment Recommendations: A Comprehensive Review

Authors: Folasade Olubusola Isinkaye, Michael Olusoji Olusanya, Pramod KumarSingh

Year: 2024

Publisher: Heliyon (Elsevier)

Summary

This paper presents a comprehensive review of deep learning and content-based filtering techniques aimed at improving plant disease identification and treatment recommendations. The authors discuss the shortcomings of traditional plant disease detection methods, which often lack accuracy, scalability, and adaptability to diverse agricultural conditions. Advanced deep learning models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Generative Adversarial Networks (GANs), and Transformer-based architectures are analyzed for their effectiveness in plant disease classification. In addition, content-based filtering techniques are explored to personalize treatment recommendations based on disease type, severity, and crop characteristics. The study also emphasizes the integration of IoT devices, remote sensing technologies, and multimodal data to enable realtime disease monitoring and intelligent decision support systems for precision agriculture.

Data Mining / Machine Learning Techniques Used

The review examines Convolutional Neural Networks (CNNs) for automated feature extraction and disease classification from plant images. Recurrent Neural Networks (RNNs) are discussed for modeling temporal patterns in disease progression. Generative Adversarial Networks (GANs) are analyzed for synthetic data generation to address dataset scarcity. Transformerbased models are reviewed for capturing complex spatial and contextual relationships. Additionally, content-based filtering techniques are employed to recommend suitable disease treatments based on historical and disease-specific data.

Research Gap Identified

The paper identifies several challenges that limit real-world deployment of AI-based plant disease detection systems. The availability of large, diverse, and real-world agricultural datasets remains limited, affecting model generalization. High computational requirements of deep learning models necessitate powerful hardware such as GPUs, increasing deployment costs. Furthermore, the black-box nature of deep learning models reduces interpretability, making it difficult for farmers and agricultural experts to trust and adopt these systems.
Challenges related to real-time monitoring and field-level implementation are also highlighted.

Future Enhancements / Scope for Improvement

Future research can focus on creating large-scale, diverse, and real-world datasets to improve the robustness and generalization of deep learning models. Developing lightweight and explainable AI models will enhance transparency and facilitate practical deployment. Integrating IoT-based sensing, multimodal data sources, and genetic information can further improve disease detection accuracy and treatment recommendations. Real-time monitoring frameworks and edge-computing solutions can also be explored to support scalable and costeffective agricultural applications.

11. Leveraging ML to predict climate change impact on rice crop disease in Eastern India

Authors: Satiprasad Sahoo, Chiranjit Singha, Ajit Govind & Mamta Sharma

Journal: Environmental Monitoring and Assessment (Springer Nature)

Year: 2025

Summary

The paper, published in the journal Environmental Monitoring and Assessment**,** investigates how climate change influences the occurrence of major rice crop diseases in Eastern India using advanced machine learning techniques. The authors integrate climatic variables (such as precipitation, maximum and minimum temperature), soil characteristics, and satellite-derived vegetation indices to model disease susceptibility under both present and future climate scenarios based on CMIP6 projections (SSP2-4.5 and SSP5-8.5). Multiple machine learning models, including Random Forest, Gradient Boosting, XGBoost, Artificial Neural Networks, and Support Vector Machines, are evaluated to identify key drivers of disease spread. The study finds that rising temperatures and changing rainfall patterns significantly increase the risk of diseases such as rice blast, brown spot, and tungro in specific districts, highlighting the growing vulnerability of rice production under climate change. Overall, the research demonstrates the usefulness of machine learning–based predictive frameworks for climate-resilient agricultural planning and sustainable crop disease management.

Data Mining / Machine Learning Techniques Used

The study employs several supervised machine learning techniques to predict the impact of climate change on rice crop disease occurrence in Eastern India. These include Random Forest (RF), Gradient Boosting Machines (GBM), XGBoost, Artificial Neural Networks (ANN), and Support Vector Machines (SVM). Data mining techniques such as feature integration and selection are applied to combine heterogeneous datasets, including climatic variables (temperature and precipitation), soil parameters, and satellite-derived vegetation indices. Ensemble learning methods, particularly Random Forest and XGBoost, are used to capture non-linear relationships between environmental factors and disease incidence. Model performance is evaluated using accuracy and other statistical metrics to identify the most influential predictors driving disease susceptibility under different climate scenarios.

Research Gap Identified

Although the study effectively demonstrates the role of machine learning in predicting climate-driven rice diseases, it primarily relies on historical and projected environmental data without incorporating real-time field-level disease observations or farmer-reported data. The models focus on district-level predictions, which may limit their applicability for plot-level disease management. Additionally, the study does not explore deep learning–based image analysis for

direct disease identification, nor does it fully address model interpretability for practical decision-making by farmers and policymakers.

Future Enhancements / Scope for Improvement

Future work can enhance the proposed framework by integrating real-time sensor data, drone or smartphone-based crop images, and Internet of Things (IoT) devices for more precise and localized disease prediction. Incorporating explainable AI techniques could improve model transparency and trust among end users. Furthermore, extending the model to include economic impact assessment and adaptive disease control recommendations would support climate-resilient agricultural planning and sustainable rice production systems.

12. An Efficient Disease Prediction in Smart Agriculture Using Advanced Deep Learning Methods for Improving Crop Productivity

Authors: Vivek Parganiha, Monika Verma

Journal: Journal of Phytopathology

Year: 2025

Summary

The paper addresses the major problem of plant diseases, which significantly reduce crop productivity worldwide, by proposing an advanced deep learning-based framework for disease prediction in smart agriculture**.** The authors develop a systematic model that integrates image preprocessing (noise removal and contrast enhancement), segmentation using a Modified Residual U-Net**,** and feature extraction via convolutional neural networks and vision transformer techniques**.** These features are then classified using an ensemble of machine learning classifiers**,** including XGBoost, Gradient Boosting, and AdaBoost-Decision Tree. Evaluated on multiple agricultural datasets, the proposed approach demonstrates very high accuracy (around 99% and above)**,** offering a robust and efficient tool for early detection of crop diseases, which can help farmers improve crop health and yield while reducing reliance on manual methods.

Data Mining / Machine Learning Techniques Used

The study employs a hybrid deep learning and ensemble learning approach for crop disease prediction. Image preprocessing techniques such as noise removal and contrast enhancement are applied to improve data quality. A Modified Residual U-Net is used for image segmentation

to accurately isolate diseased regions. For feature extraction, the authors utilize Convolutional Neural Networks (CNNs) combined with Vision Transformer (ViT) architectures to capture both local and global image features.

Research Gap Identified

Although the proposed framework achieves very high accuracy, the study primarily focuses on controlled and benchmark datasets, which may not fully represent real-world agricultural environments. The model's performance under varying field conditions such as changing lighting, background noise, occlusions, and different camera qualities is not extensively evaluated. Additionally, the framework requires high computational resources due to the use of deep learning and ensemble models, which limits its feasibility for deployment on low-cost or edge-based agricultural devices commonly used by small-scale farmers.

Future Enhancements / Scope for Improvement

Future work can focus on optimizing the model for real-time and edge-based deployment by reducing computational complexity through model compression and lightweight architectures. Expanding the dataset to include real-field images captured under diverse environmental conditions would improve generalization and robustness. Integration with IoT sensors and mobile applications can further enhance practical usability, enabling farmers to receive instant disease diagnosis and actionable recommendations for crop management.

13. Predicting crop disease severity using real time weather variability through machine learning algorithms

Authors: Amit Bijlwan, Rajeev Ranjan, Manendra Singh, Rahul Purohit, Arun Jyoti Nath & Sumit Chakravarty

Journal: Scientific Reports

Year: 2025

Summary

The study "Predicting crop disease severity using real time weather variability through machine learning algorithms" investigates how integrating real-time meteorological data with advanced machine learning models can accurately predict the severity of two major wheat diseases, yellow rust and powdery mildew. Conducted over two growing seasons in India, the research combined weekly disease assessments with weather variables (e.g., temperature, humidity,

rainfall, wind speed) and compared multiple models such as artificial neural networks (ANN) and random forest**.** The ANN model performed best, achieving high predictive accuracy (with R^2 values up to ~0.98 in calibration and ~0.95 in validation), while other models also showed strong performance. Principal component analysis identified key weather factors driving disease severity, highlighting the importance of real-time weather data for forecasting. The findings suggest that machine learning–based disease prediction systems could serve as valuable tools for early warning and improved crop management, helping farmers anticipate outbreaks and reduce yield losses.

Data Mining / Machine Learning Techniques Used

The study employed several machine learning and data mining techniques to predict crop disease severity using real-time weather data. Artificial Neural Networks (ANN) and Random Forest (RF) models were primarily used to capture complex, non-linear relationships between meteorological variables and disease severity. Principal Component Analysis (PCA) was applied as a dimensionality reduction technique to identify the most influential weather factors affecting disease progression. Model performance was evaluated using statistical metrics such as R^2 values during calibration and validation phases, ensuring robustness and accuracy of predictions.

Research Gap Identified

Although the study demonstrates high predictive accuracy using ANN and RF models, it focuses only on two wheat diseases (yellow rust and powdery mildew) and is limited to specific geographic regions and seasonal conditions in India. The model does not incorporate crop image data, soil health parameters, or pest-related factors, which could further influence disease severity. Additionally, real-time deployment challenges such as scalability, sensor reliability, and farmer-level accessibility were not addressed.

Future Enhancements / Scope for Improvement

Future work can extend this approach to multiple crops and disease types across diverse agro-climatic regions. Integrating deep learning models, satellite imagery, IoT-based sensors, and soil nutrient data could further enhance prediction accuracy. Developing mobile or web-based decision support systems with real-time alerts would improve practical usability for farmers.

Moreover, adaptive and self-learning models that update continuously with new weather and disease data can make the system more resilient to climate variability.

14. AI-Driven Crop Disease Prediction and Management System

Authors: Shishir Shastry BH; Aryaman Chakraborty; Abhiram Palukuru; Aishwarya M; Dr. Asha PN

Journal: Iconic Research And Engineering Journals (IRE Journals)

Year: 2025

Summary

The paper *"AI-Driven Crop Disease Prediction and Management System"* presents a deep learning-based approach to detect and manage crop diseases to improve agricultural productivity. It highlights that traditional disease identification methods are often slow and unreliable, so the authors propose an AI system using Convolutional Neural Networks (CNNs) to analyze leaf images of crops such as apple, grape, corn, potato, and tomato for early disease detection. The model's performance is compared with transfer learning approaches like VGG16, and the system not only identifies diseases accurately but also suggests management strategies to help reduce crop loss and enhance yield, making it useful for farmers and agricultural research.

Data Mining / Machine Learning Techniques Used

The paper employs Deep Learning techniques, primarily Convolutional Neural Networks **(CNNs)**, for automatic crop disease detection from leaf images. Image preprocessing and feature extraction are handled within the CNN architecture. The study also compares the proposed CNN model with transfer learning models such as VGG16 to evaluate performance improvements. Classification is performed based on visual disease patterns learned from the image dataset, enabling accurate multi-class disease prediction.

Research Gap Identified

Although the proposed system demonstrates high accuracy in disease detection, the study does not extensively address model generalization across diverse real-world conditions**,** such as varying lighting, backgrounds, and image quality. Additionally, the system focuses mainly on disease identification and basic management suggestions, without integrating real-time field

data, environmental factors, or severity estimation, which limits its practical deployment at scale.

15. AI-IoT based smart agriculture pivot for plant diseases detection and treatment

Authors: Amin S. Ibrahim, Saeed Mohsen, I. M. Selim, Roobaea Alroobaea, Majed Alsafyani, Abdullah M. Baqasah & Mohamed Eassa

Journal: Scientific Reports (Nature Publishing Group)

Year: 2025

Summary

This paper proposes an AI-IoT-based smart agriculture pivot system designed to detect and treat plant diseases more efficiently than traditional drones or robotics. By integrating artificial intelligence (AI) with Internet of Things (IoT) technologies on a modified central pivot irrigation system, the authors build a hardware and software platform that captures plant leaf images, processes them with deep learning (ResNet50) to classify health status across 11 classes (including healthy and diseased crops), and automatically actuates fertilization or irrigation responses when disease is detected. The system achieved very high performance in disease classification (about 99.8% accuracy, with F1-score, recall, and precision also near 100%), and includes a mobile app to help farmers identify plant health and receive treatment guidance, making it a practical tool for boosting crop productivity and disease management in smart farming.

Data Mining / Machine Learning Techniques Used

The study employs deep learning techniques, specifically ResNet50, to classify plant leaf images into 11 classes, including healthy and diseased crops. Image data collected via IoT sensors is preprocessed and fed into the model for feature extraction and classification. The integration of AI with IoT enables real-time analysis and automated response actions, bridging data acquisition and actionable decision-making in smart agriculture.

Research Gap Identified

Traditional disease detection systems using drones or robotics often lack real-time monitoring, require manual intervention, and may not scale efficiently for large farms. Existing IoT solutions rarely integrate automated treatment mechanisms alongside disease detection, and

most fail to achieve high accuracy across multiple crop types. This study addresses the gap by combining AI-based disease detection with IoT-enabled automated irrigation and fertilization.

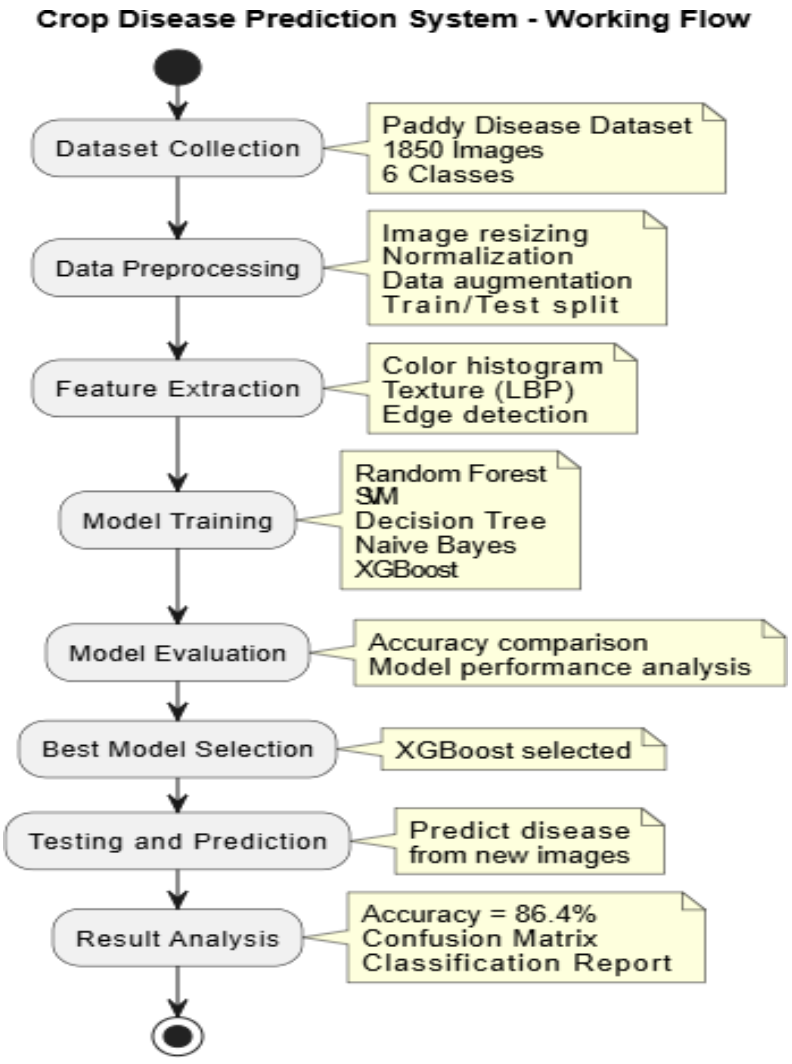
Future Enhancements / Scope for Improvement

Future improvements could include expanding the system to cover a wider range of crops and disease types, integrating predictive analytics for disease outbreaks, and optimizing the hardware for energy efficiency and cost-effectiveness. Additionally, incorporating edge computing could reduce latency in real-time processing, and advanced deep learning models like vision transformers (ViTs) could further improve classification accuracy and adaptability to varying environmental conditions.

4. METHODOLOGY

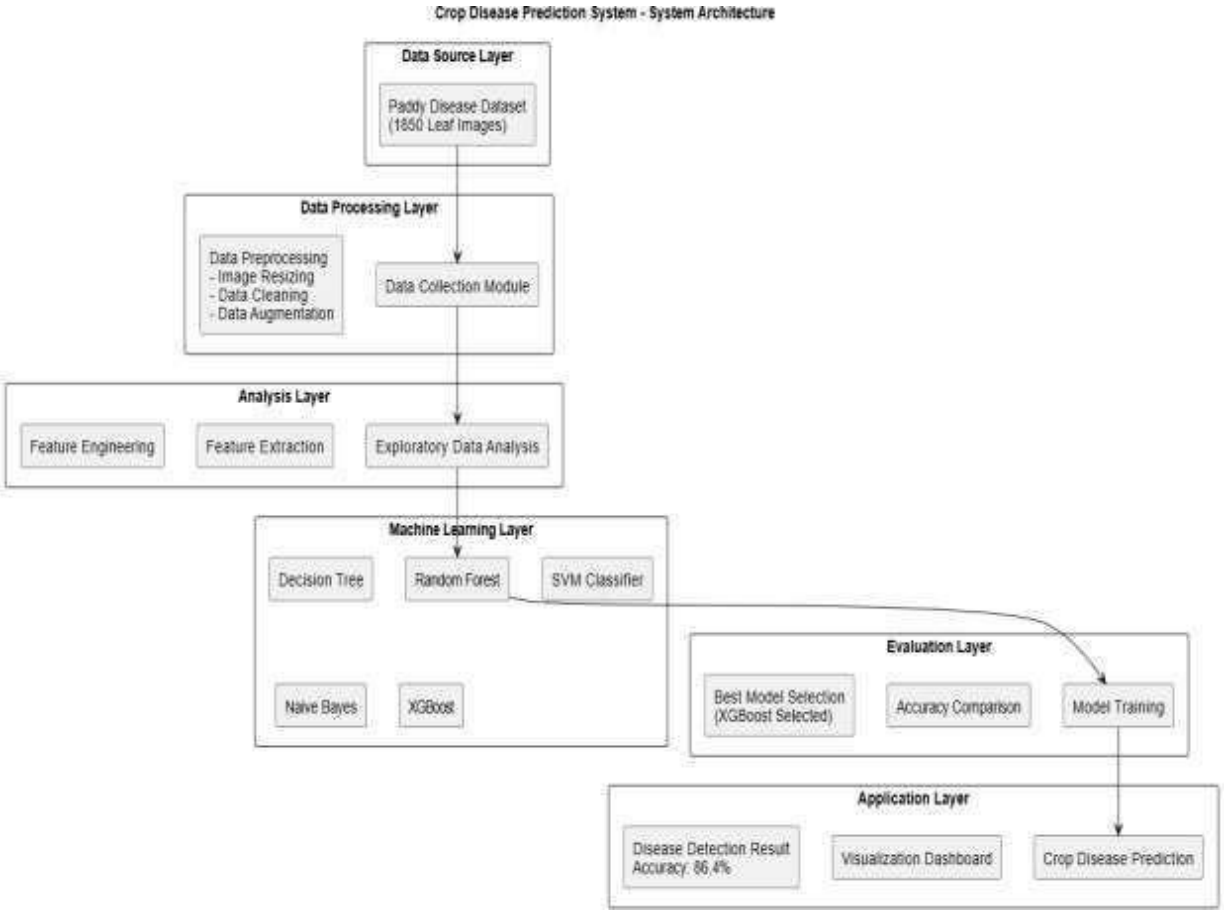
4.1 Flowchart

Below diagram is Flowchart which explains working flow of our Proposed System.



4.2 System Architecture Diagram:

Below Diagram is our Proposed System Architecture Diagram:



4.3 Pseudocode:

Input: Paddy leaf image dataset and new leaf image for testing.

1. Load the Paddy Leaf Disease image dataset.

2. For each image in the dataset
   Read the image
   Resize image to fixed size (100 × 100)
   Convert image to RGB format
   Store image data and corresponding class label
   END FOR

3. Extract Features from each image
   a. Color Features
      Calculate mean and standard deviation of R, G, B channels
   b. Texture Features
      Convert image to grayscale
      Compute GLCM matrix
      Extract contrast, homogeneity, energy, correlation
   c. Shape Features
      Apply thresholding
      Detect contours
      Calculate area and perimeter

4. Store all extracted features into a feature dataset.

5. Encode class labels using Label Encoder.

6. Split dataset into Training Set and Testing Set
   Training data = 80%
   Testing data = 20%

7. Normalize features using Standard Scaler.

8. Initialize Machine Learning Models
   Logistic Regression
   Decision Tree
   Random Forest
   Support Vector Machine
   k-Nearest Neighbors
   Naive Bayes
   XGBoost

9. For each model
   Train the model using training data
   Predict results using testing data
   Calculate accuracy and training time
   END FOR

10. Compare performance of all models.

11. Select the best performing model.

12. Evaluate the best model
    Generate confusion matrix
    Generate classification report

13. Display model accuracy and evaluation results.

Output: Predicted crop disease type and model accuracy using the XGBoost classifier.

4.4 Code:

Dataset preprocessing Code:

Step 1: Importing Necessary Libraries

```
# Install required packages if needed
!pip install scikit-image opencv-python matplotlib numpy pandas

import os
import numpy as np
import pandas as pd
from skimage.io import imread
from skimage.transform import resize
import matplotlib.pyplot as plt
%matplotlib inline
```

```
# 1. Define Dataset Path (using raw string for Windows paths)
dataset_path = r'C:\Users\yogas\OneDrive\Desktop\Paddy Leaf Disease'
categories = ['Bacterial_leaf_blight', 'Brown_spot', 'Healthy_leaf',
              'Leaf_blast', 'Other', 'Tungro']

# 2. Initialize Data Storage
flat_data = []    # For flattened image arrays
target = []       # For category labels
image_paths = []  # To keep track of source images

# 3. Set Image Parameters
img_width, img_height = 100, 100  # Standard size for all images
channels = 3  # For RGB images
```

Step 2: Load the Dataset

```
from google.colab import files
uploaded = files.upload()
```

```
Choose Files  Paddy Leaf Disease.zip
Paddy Leaf Disease.zip(application/x-zip-compressed) - 32286233 bytes, last modified: 11/03/2026 - 100% done
Saving Paddy Leaf Disease.zip to Paddy Leaf Disease.zip
```

Step 3: Initialize Data Storage

```
flat_data=[]
target=[]
image_paths=[]

img_size=100

for category in categories:
    category_path=os.path.join(dataset_path,category)

    for img in os.listdir(category_path):
        img_path=os.path.join(category_path,img)

        image=imread(img_path)
        image_resized=resize(image,(img_size,img_size,3))

        flat_data.append(image_resized.flatten())
        target.append(category)
        image_paths.append(img_path)
```

Step 4: Load and Resize Images

```
from skimage.io import imread
from skimage.transform import resize
import numpy as np
import pandas as pd
import os

dataset_path = "dataset/Paddy Leaf Disease"

categories = os.listdir(dataset_path)

flat_data = []
target = []
image_paths = []

img_size = 100

for category in categories:

    category_path = os.path.join(dataset_path, category)
```

```
for category in categories:

    category_path = os.path.join(dataset_path, category)

    for img in os.listdir(category_path):

        img_path = os.path.join(category_path, img)

        try:
            image = imread(img_path)
            image_resized = resize(image, (img_size, img_size, 3))

            flat_data.append(image_resized.flatten())
            target.append(category)
            image_paths.append(img_path)

        except:
            continue
```

Step 5: Create DataFrame from Image Data

```
        except:
            continue

df = pd.DataFrame(flat_data)
df['Target'] = target
df['ImagePath'] = image_paths

print("Total images loaded:", len(df))
```

```
Total images loaded: 1858
```

Step 6: Display Sample Images from Dataset

```
plt.figure(figsize=(15,10))

for i, category in enumerate(categories):

    sample = df[df['Target'] == category].iloc[0]

    img_array = sample.drop(['Target','ImagePath']).values.astype(float)

    img = img_array.reshape(100,100,3)

    # Convert to proper range if needed
    if img.max() <= 1:
        img = (img * 255).astype(np.uint8)

    plt.subplot(2,3,i+1)
    plt.imshow(img)
    plt.title(category)
    plt.axis('off')
```

```
plt.tight_layout()
plt.show()
```



Step 7: Split Dataset into Training and Testing Sets

```
# Encode labels
label_encoder = LabelEncoder()
y_encoded = label_encoder.fit_transform(y)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y_encoded,
    test_size=0.2,
    random_state=42,
    stratify=y_encoded
)

print("Training set:", X_train.shape)
print("Testing set:", X_test.shape)
```

```
Training set: (1480, 30000)
Testing set: (370, 30000)
```

Step 8: Dimensionality Reduction using PCA

```
from sklearn.decomposition import PCA

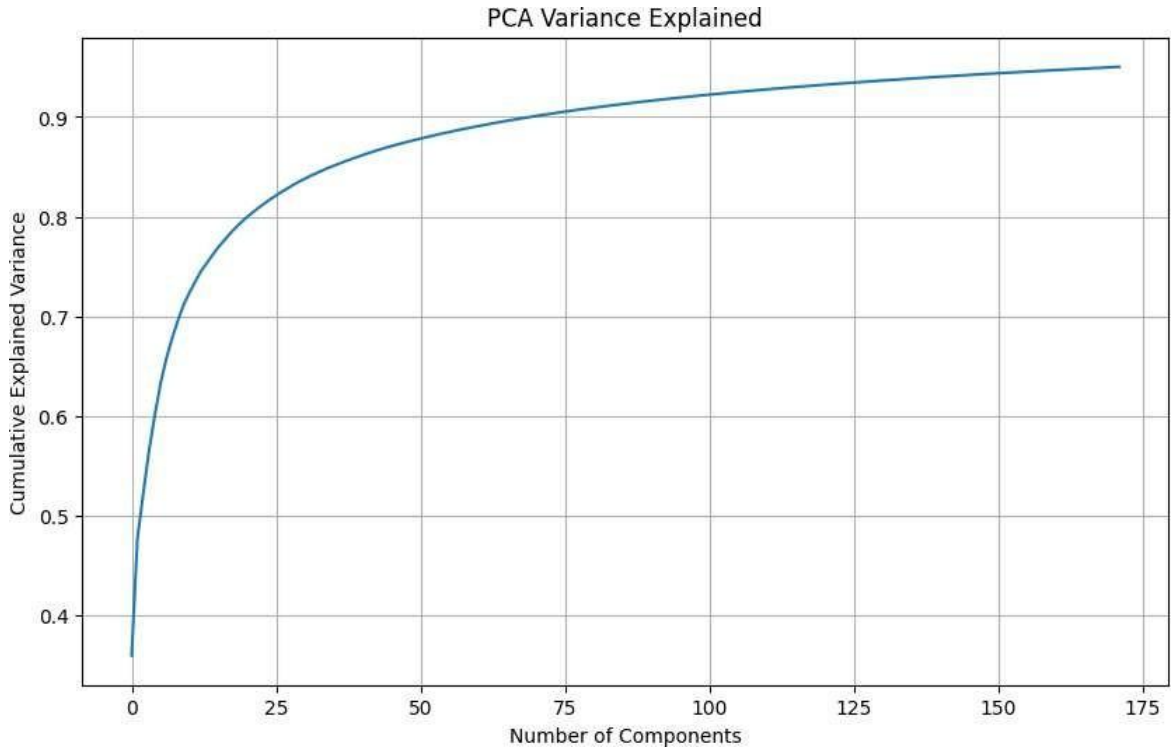
# Apply PCA to reduce dimensions while preserving 95% variance
pca = PCA(n_components=0.95, random_state=42)
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)

print(f"Original dimensions: {X_train_scaled.shape[1]}")
print(f"Reduced dimensions: {X_train_pca.shape[1]}")
print(f"Variance retained: {sum(pca.explained_variance_ratio_):.2%}")

# Visualize the explained variance
plt.figure(figsize=(10,6))
plt.plot(np.cumsum(pca.explained_variance_ratio_))
plt.xlabel('Number of Components')
plt.ylabel('Cumulative Explained Variance')
plt.title('PCA Variance Explained')
plt.grid()
plt.show()
```

```
plt.ylabel('Cumulative Explained Variance')
plt.title('PCA Variance Explained')
plt.grid()
plt.show()
```

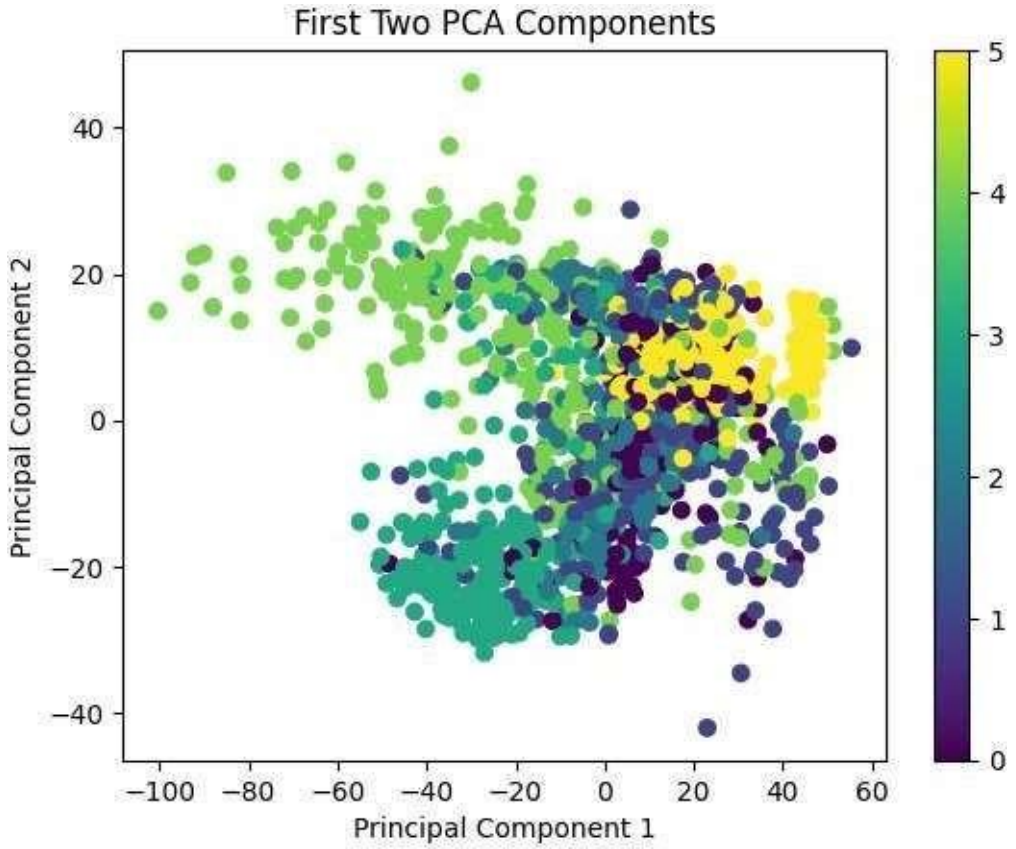
```
Original dimensions: 30000
Reduced dimensions: 172
Variance retained: 95.02%
```



Step 9 :First Two PCA Components Scatter Plot

```
pca_2 = PCA(n_components=2)
X_pca_2 = pca_2.fit_transform(X_train_scaled)

plt.figure()
plt.scatter(X_pca_2[:,0], X_pca_2[:,1], c=y_train, cmap='viridis')
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("First Two PCA Components")
plt.colorbar()
plt.show()
```



Step 10: Feature Extraction

```
import cv2
import numpy as np
from skimage.feature import graycomatrix, graycoprops
from skimage.color import rgb2gray
import matplotlib.pyplot as plt
%matplotlib inline

# Function to extract multiple features from an image
def extract_features(image):
    features = {}

    # Convert to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_RGB2GRAY)

    # 1. Color Features (Mean and Std Dev)
    for i, channel in enumerate(['r', 'g', 'b']):
        features[f'{channel}_mean'] = np.mean(image[:,:,i])
        features[f'{channel}_std'] = np.std(image[:,:,i])
```

```
    # 2. Texture Features (GLCM)
    glcm = graycomatrix(gray, distances=[1], angles=[0], levels=256, symmetric=True, normed=True)
    for prop in ['contrast', 'dissimilarity', 'homogeneity', 'energy', 'correlation']:
        features[prop] = graycoprops(glcm, prop)[0, 0]

    # 3. Shape Features
    _, thresh = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)
    contours, _ = cv2.findContours(thresh, cv2.RETR_TREE, cv2.CHAIN_APPROX_SIMPLE)
    if len(contours) > 0:
        cnt = max(contours, key=cv2.contourArea)
        features['area'] = cv2.contourArea(cnt)
        features['perimeter'] = cv2.arcLength(cnt, True)
    else:
        features['area'] = 0
        features['perimeter'] = 0

    return features
```

```
# Extract features from sample images
sample_features = []
for i in range(6):  # For each class
    sample = df[df['Target'] == categories[i]].iloc[0]
    img_array = sample.drop(['Target', 'ImagePath']).values
    img = img_array.reshape(img_width, img_height, channels)
    if img.max() <= 1.0:
        img = (img * 255).astype(np.uint8)
    features = extract_features(img)
    features['class'] = categories[i]
    sample_features.append(features)

# Convert to DataFrame
features_df = pd.DataFrame(sample_features)
print("\nExtracted Features:")
display(features_df)
```

Extracted Features:

	r_mean	r_std	g_mean	g_std	b_mean	b_std	contrast	dissimilarity	homogeneity	energy	correlation	area	perimeter	
0	114.8909	63.440631	125.6585	63.822746	128.6318	82.258120	133.074444	4.299091	0.470203	0.073590	0.984650	3334.5	277.355338	Le
1	211.2876	61.172104	216.5599	55.645375	200.8712	82.393421	318.488889	4.738182	0.621006	0.168772	0.956331	4749.5	326.095452	Bro
2	91.0003	36.442921	89.7889	36.734816	68.0514	35.820485	422.404747	14.756869	0.084185	0.014197	0.831048	170.5	152.610171	
3	179.9809	41.327857	185.6183	36.593800	181.1538	62.451876	140.911818	3.065152	0.630839	0.110772	0.955868	5586.5	322.870057	Heal
4	205.0280	27.031001	205.3964	26.886251	194.6410	62.727279	12.135152	1.014545	0.671561	0.125547	0.993360	9260.0	546.426408	
5	169.4765	68.327162	167.4499	56.534022	127.1570	89.892391	224.684141	5.994040	0.416266	0.099304	0.972272	3328.0	265.656854	Bacterial_le

Step 11: Feature Visualization of Extracted Color, Texture, and Shape Features

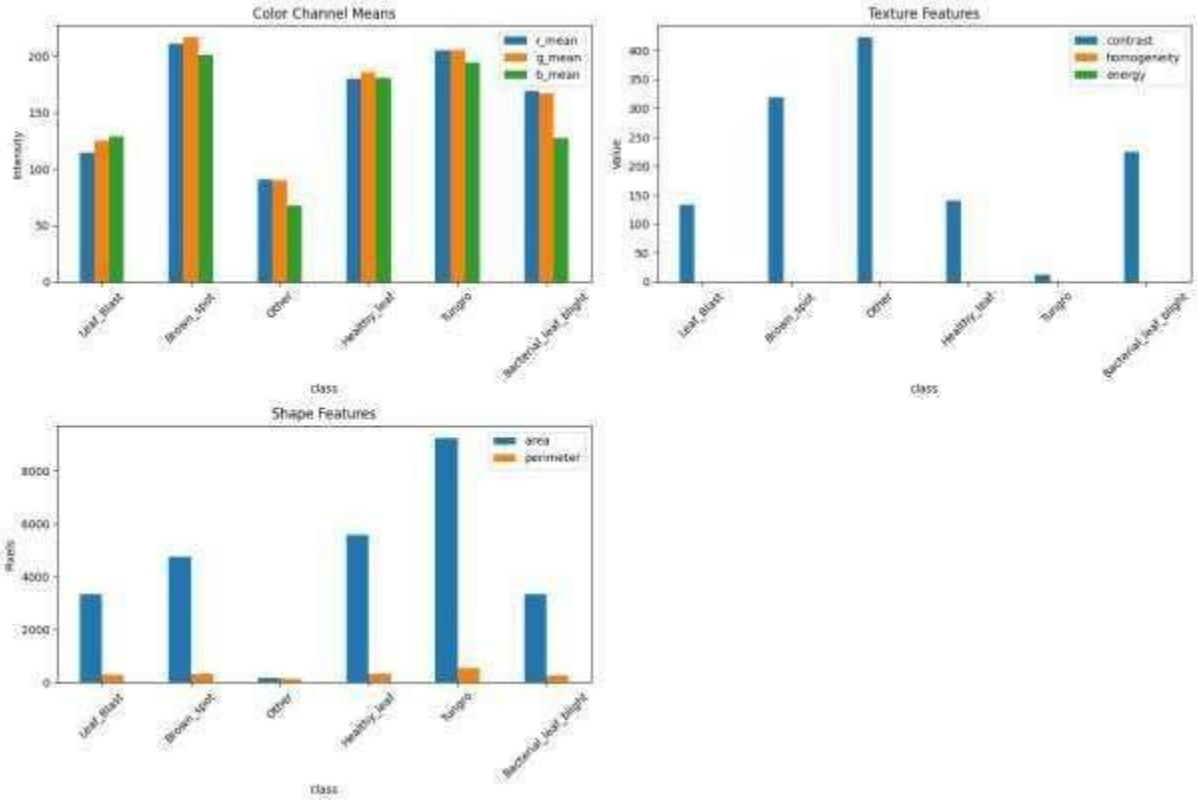
```
    # Visualize important features
    plt.figure(figsize=(15, 10))

    # 1. Color Features
    plt.subplot(2, 2, 1)
    features_df.plot(x='class', y=['r_mean', 'g_mean', 'b_mean'], kind='bar', ax=plt.gca())
    plt.title('Color Channel Means')
    plt.ylabel('Intensity')
    plt.xticks(rotation=45)

    # 2. Texture Features
    plt.subplot(2, 2, 2)
    features_df.plot(x='class', y=['contrast', 'homogeneity', 'energy'], kind='bar', ax=plt.gca())
    plt.title('Texture Features')
    plt.ylabel('Value')
    plt.xticks(rotation=45)
```

```
    # 3. Shape Features
    plt.subplot(2, 2, 3)
    features_df.plot(x='class', y=['area', 'perimeter'], kind='bar', ax=plt.gca())
    plt.title('Shape Features')
    plt.ylabel('Pixels')
    plt.xticks(rotation=45)

    plt.tight_layout()
    plt.show()
```



Step12 : Feature Extraction from All Paddy Leaf Images

```
# Extract features from all images (this may take time)
all_features = []

for idx, row in df.iterrows():
    img_array = row.drop(['Target', 'ImagePath']).values
    img = img_array.reshape(img_width, img_height, channels)
    if img.max() <= 1.0:
        img = (img * 255).astype(np.uint8)
    features = extract_features(img)
    features['class'] = row['Target']
    all_features.append(features)

# Convert to DataFrame
all_features_df = pd.DataFrame(all_features)

# Save extracted features
all_features_df.to_csv('extracted_features.csv', index=False)
print("Feature extraction completed and saved to extracted_features.csv")
```

```
Feature extraction completed and saved to extracted_features.csv
```

Step 13: Visualization of Image Samples with Extracted Texture Features



Step 14: Model Training and Performance Evaluation

```
# Prepare data (assuming features_df exists from previous steps)
if 'features_df' not in locals():
    features_df = pd.read_csv('extracted_features.csv')

# Encode labels
le = LabelEncoder()
y = le.fit_transform(features_df['class'])
X = features_df.drop('class', axis=1)

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

# Scale features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
# Train and evaluate models
results = []
for name, model in models.items():
    try:
        start_time = time.time()

        # Train model
        model.fit(X_train_scaled, y_train)
        train_time = time.time() - start_time

        # Predictions
        y_pred = model.predict(X_test_scaled)
        y_pred_train = model.predict(X_train_scaled)

        # Calculate metrics
        test_accuracy = accuracy_score(y_test, y_pred)
        train_accuracy = accuracy_score(y_train, y_pred_train)

        results.append({
            'Model': name,
            'Train Accuracy': train_accuracy,
```

```
    except Exception as e:
        print(f"Error with {name}: {str(e)}")
        continue

# Convert results to DataFrame
results_df = pd.DataFrame(results)
if not results_df.empty:
    print("\nModel Performance Comparison:")
    display(results_df.sort_values(by='Test Accuracy', ascending=False))
else:
    print("No models were successfully trained")
```

```
Logistic Regression completed successfully
Decision Tree completed successfully
Random Forest completed successfully
Gradient Boosting completed successfully
SVM completed successfully
k-NN completed successfully
Naive Bayes completed successfully
XGBoost completed successfully
```

Model Performance Comparison:

	Model	Train Accuracy	Test Accuracy	Training Time (s)
7	XGBoost	1.000000	0.864865	5.697072
4	SVM	0.831757	0.840541	0.114419
3	Gradient Boosting	0.996622	0.837838	14.603238
2	Random Forest	1.000000	0.837838	1.204456
5	k-NN	0.857432	0.824324	0.026088
0	Logistic Regression	0.825000	0.813514	0.261912
1	Decision Tree	1.000000	0.716216	0.056442
6	Naive Bayes	0.626351	0.600000	0.006592

Step 15: Model Performance Visualization and Result Analysis

```
if not results_df.empty:
    plt.figure(figsize=(14, 6))

    # Accuracy comparison
    plt.subplot(1, 2, 1)
    sns.barplot(x='Test Accuracy', y='Model',
                data=results_df.sort_values('Test Accuracy'))
    plt.title('Model Accuracy Comparison')
    plt.xlim(0, 1)

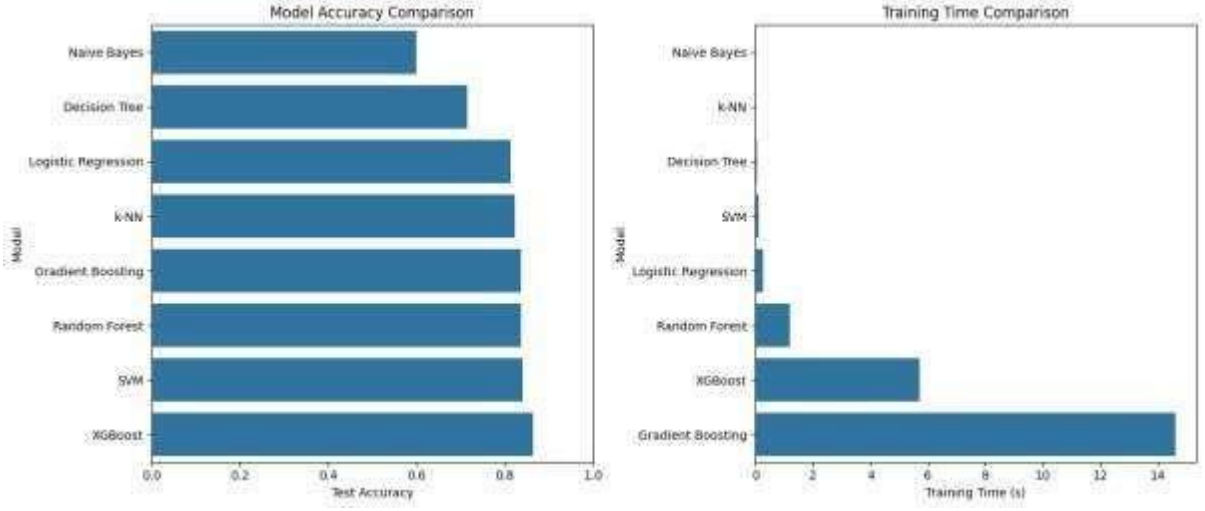
    # Training time comparison
    plt.subplot(1, 2, 2)
    sns.barplot(x='Training Time (s)', y='Model',
                data=results_df.sort_values('Training Time (s)'))
    plt.title('Training Time Comparison')
    plt.tight_layout()
    plt.show()
```

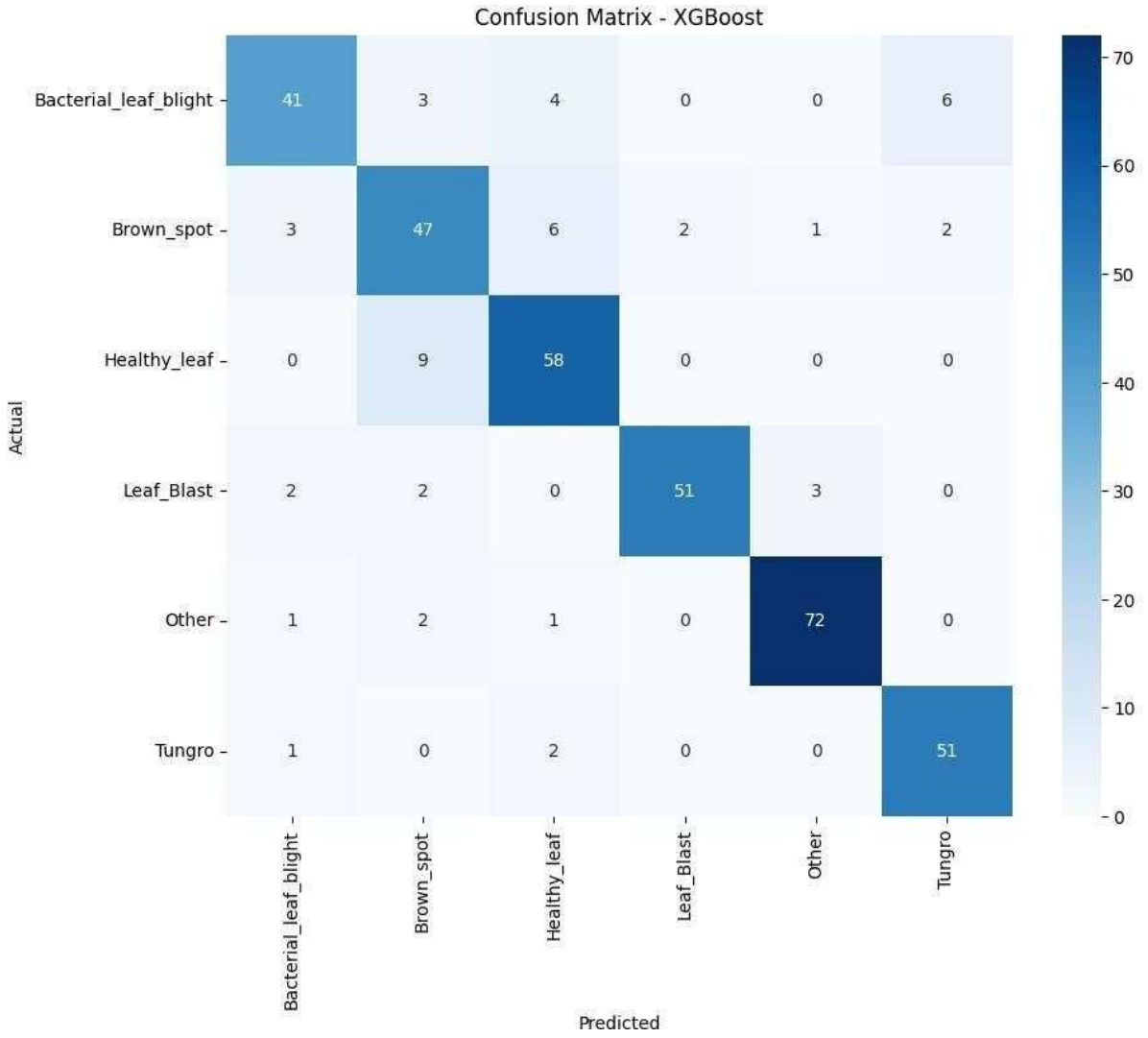
```
    plt.show()

    # Detailed report for best model
    best_model_name = results_df.loc[results_df['Test Accuracy'].idxmax(), 'Model']
    best_model = models[best_model_name]
    y_pred = best_model.predict(X_test_scaled)

    # Confusion matrix
    plt.figure(figsize=(10, 8))
    cm = confusion_matrix(y_test, y_pred)
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
                xticklabels=le.classes_, yticklabels=le.classes_)
    plt.title(f'Confusion Matrix - {best_model_name}')
    plt.xlabel('Predicted')
    plt.ylabel('Actual')
    plt.show()

    # Classification report
    print(f"\nClassification Report for {best_model_name}:")
    print(classification_report(y_test, y_pred, target_names=le.classes_))
```





Classification Report for XGBoost:

	precision	recall	f1-score	support
Bacterial_leaf_blight	0.85	0.76	0.80	54
Brown_spot	0.75	0.77	0.76	61
Healthy_leaf	0.82	0.87	0.84	67
Leaf_Blast	0.96	0.88	0.92	58
Other	0.95	0.95	0.95	76
Tungro	0.86	0.94	0.90	54
accuracy			0.86	370
macro avg	0.87	0.86	0.86	370
weighted avg	0.87	0.86	0.86	370

5. EXPERIMENTATION

The experimentation phase involved training and testing the crop disease prediction system using the Paddy Disease Dataset. The dataset consists of 1,850 labeled paddy leaf images belonging to six different classes.

The images were first preprocessed to improve model performance. Each image was resized to 100 × 100 pixels, converted into RGB format, and normalized. Important image features such as color, texture, and shape were then extracted for classification.

The dataset was divided as follows:

- Training data – 80%
- Testing data – 20%

The following disease classes were included:

- Bacterial Leaf Blight
- Brown Spot
- Healthy Leaf
- Leaf Blast
- Others
- Tungro

During experimentation, different machine learning models were trained and compared:

- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- Naive Bayes
- k-Nearest Neighbors (k-NN)
- XGBoost

The work was carried out using the following hardware and software:

Hardware

- Intel Core i5 / i7 processor
- 8 GB RAM
- 256 GB storage

Software

- Python
- Jupyter Notebook / Google Colab
- OpenCV
- Scikit-learn
- XGBoost
- NumPy and Pandas

This experimentation process helped in identifying the best model for accurate crop disease prediction.

6. RESULT & ANALYSIS

The proposed crop disease prediction system was tested using multiple machine learning algorithms to identify the most accurate model for paddy leaf disease classification. The models were trained and evaluated using the preprocessed dataset, and their performance was compared based on standard evaluation metrics.

After analysis, XGBoost produced the best results among all the models tested, with an overall accuracy of 86.4%. This shows that the model is capable of effectively predicting different paddy leaf diseases from image features.

The performance was analyzed using the following metrics:

- Accuracy: 86.4%
- Precision: High for most disease classes
- Recall: Good detection rate for infected leaves
- F1-score: Balanced model performance
- Confusion Matrix: Used to compare actual and predicted classes

The analysis also showed that:

- Healthy leaves were identified accurately
- Brown Spot and Leaf Blast were predicted effectively
- A few misclassifications occurred between visually similar diseases
- Feature extraction improved the model's classification performance

Overall, the results indicate that the proposed system is reliable for early disease detection and can support farmers in taking timely action for crop protection.

7. CONCLUSION

This project successfully developed a **crop disease prediction system** using machine learning techniques for paddy leaf disease classification. The main objective was to identify crop diseases at an early stage through image-based analysis, helping to reduce crop loss and improve agricultural productivity. The model was trained using the **Paddy Disease Dataset with 1,850 labeled images** divided into six disease categories.

Different machine learning algorithms such as Decision Tree, Random Forest, Support Vector Machine, Naive Bayes, k-NN, and XGBoost were implemented and compared. Among them, **XGBoost achieved the highest accuracy of 86.4%**, making it the best model for this project. Preprocessing steps like image resizing, normalization, and feature extraction improved the performance of the model.

The results prove that the system can effectively support early disease detection and reduce dependence on manual inspection. This helps farmers take timely preventive actions and improve crop management.

In future, the system can be enhanced using deep learning techniques, IoT-based monitoring, and mobile applications for real-time disease prediction and better farmer support.